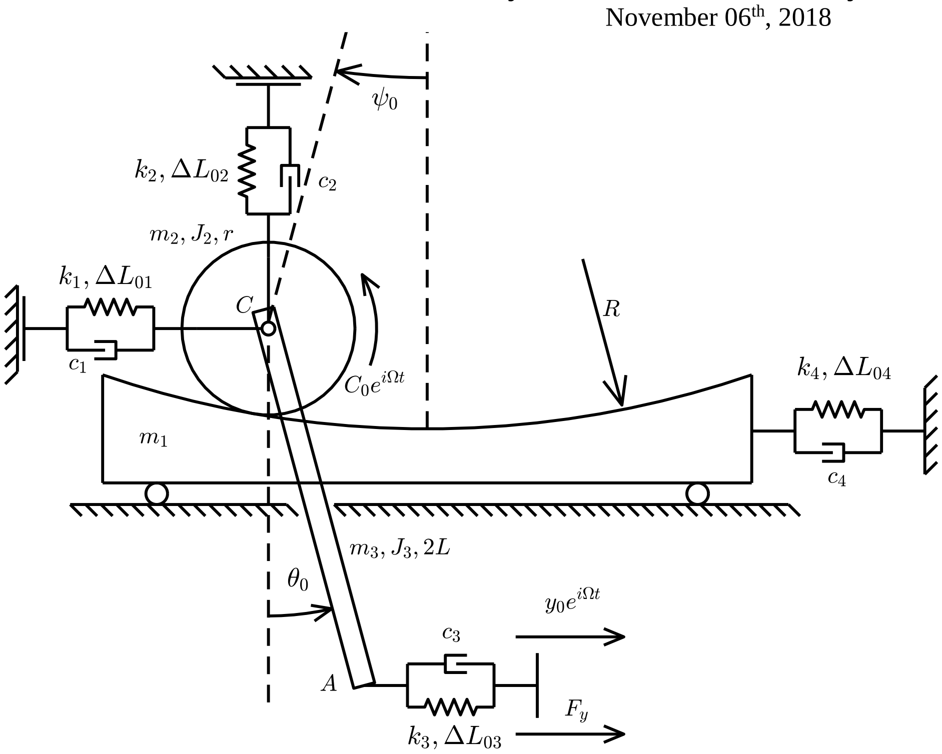
Dynamics of Mechanical Systems

November 06th, 2018



m_1 = 15 kg	k_1 = 6000 N/m
m_2 = 5 kg	k_2 = 6000 N/m
m_3 = 5 kg	k_3 = 6000 N/m
J_2 = 0.1 kgm^2	k_4 = 6000 N/m
J_3 = 0.15 kgm^2	c_1 = 2 Ns/m
r = 0.2 m	c_2 = 2 Ns/m
R = 1.0 m	c_3 = 4 Ns/m
L = 0.3 m	c_4 = 4 Ns/m
ΔL_{01} = 0.03 m	$\theta_0 = \pi/6$
ΔL_{02} = -0.0868 m	$\psi_0 = \pi/12$
ΔL_{03} = 0.0024 m	
ΔL_{04} = 0.0276 m	
ΔL_{01_new} = 0.015 m	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the upper circular part of the cart m_1, whose radius is R and whose center is O (not indicated in the drawing). In the equilibrium position the angle between the vertical direction and line OC is ψ_0. The upper extremity of the rigid bar AC is connected to the disk's center through a hinge, and its lower part is connected to the ground through a spring and damper system (k_3, c_3). A horizontal constraint displacement is applied to the left extremity of the spring and damper system (k_3, c_3). In the equilibrium position the angle between the vertical direction and the AC bar is θ_0.

The students are requested to:

1. Write their surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "*Nr.*" Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the displacement of the cart m_1 and the horizontal component of the displacement of point A. Save the graphical results as SNxxx1.fig and SNxxx2.fig, whit S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the vertical component of the displacement of point C and the constraint force F_y in the moving constraint to which the spring/damper system (k_3, c_3) is attached. Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the horizontal dynamic component of the constraint force in the hinge C and the tangential component of the contact force between disk and cart m_1. Save the graphical result as SNxxx5.fig and SNxxx6.fig;
7. Compute the static preload of springs k_2, k_3 and k_4 keeping the same static configuration and changing the static preload of spring k_1 from ΔL_{01} to ΔL_{01_new}.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site https://beep.metid.polimi.it and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test November 06th 2018 B6.3.2* or *Test November 06th 2018 B6.2.8* or *Test November 06th 2018 B8.2.2* according to your classroom
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse… (or Sfoglia…)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.